Daniel Jea

(845) 594-3655 | Boston, MA | Availability: Jan – Aug 2027
jea.d@northeastern.edu | github.com/agrvspr | linkedin.com/in/daniel-jea

Education

Northeastern University Boston, MA
Bachelor of Science in Computer Science *May 2028*

- Honors: GPA: 3.7/4, Dean's List (2025 – Present)
- Related Courses: Fundamentals of Computer Science 1 & 2, Object Oriented Design, Algorithms and Data, Discrete Structures, Foundations of Cybersecurity, Foundations of Data Science, Computer Systems, Web Development, Logic and Computation, Digital Design and Computer Architecture, Principles of Artificial Intelligence, Network Fundamentals, Foundations of Software Engineering, Programming Languages, Intro to Databases

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, OCaml, Racket, SQL
Frameworks/Libraries: React, Next.js, Vercel, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
Tools: GitHub, Git, VS Code, IntelliJ IDEA

Extracurricular Experience

CTF (Capture the Flag) Club | *Member* Jan. 2026 – Present

- Compete in club-organized Capture the Flag competitions, applying skills in OSINT, cryptography, and forensics

AINU (Artificial Intelligence Club) | *Member* Jan. 2025 – Present

- Participated in AI-focused bootcamps and workshops covering emerging AI technologies, tools, and practical applications.
- Engaged in discussions and collaborative activities exploring real-world uses of AI across academic, professional, and everyday settings.

Project Experience

Cat and Mouse | *Python* July 2026 – Aug. 2026

- Co-authored research on a dual-brain enemy AI combining A* pathfinding with Q-learning reinforcement learning;
- Assisted in the creation of a generative algorithm to create the levels
- Proposed and helped implement a tick-based enemy speed system, decoupling enemy movement cadence from player input frequency to create smoother, more consistent difficulty scaling

Screen Translator | *Python* Jul. 2026 – Present

- Directed AI-assisted development of a real-time on-screen OCR translation tool (Korean/Chinese/Japanese to English) with an MVC-style architecture, resolving a PaddlePaddle framework regression and Python-version dependency conflicts to enable reliable local inference
- Designed and validated automatic language detection by scoring OCR confidence across 3 candidate recognition models, achieving 100% accuracy on clean digital text and 80% on low-resolution, mixed-language cases via a custom Python test harness across 60 labeled images

What's That Pokémon? | *Python* Sept. 2025 – Dec. 2025

- Implemented a machine learning algorithm to predict Gen 1 Pokémon's type through battle statistics (41.30% first-guess accuracy, 76.09% top-three guess accuracy)
- Built 10+ visualizations using Scikit-learn and Seaborn to display average statistics per Pokémon type
- Designed a dual-type prediction system that evaluates top 3 predictions against both primary and secondary Pokémon types, better reflecting the multi-label nature of the problem

Kambaz | *TypeScript* Jan. 2026 – May 2026

- Developed a multi-page web application using Next.js, React, and TypeScript, implementing dynamic routing and reusable UI components
- Created reusable layout and navigation components to ensure consistent UI and improve code maintainability
- Designed a Canvas-inspired learning management interface, focusing on intuitive navigation and user experience
- Developed a Node.js backend server to handle API requests and connect the frontend to application data
- Deployed the frontend on Vercel, configuring builds and routing for production